

Active Inference BookStream #001.06 ~ "Governing Continuous Transformation"

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hello and welcome this is active bookstream number 001.06 on governing continuous transformation it's January 17 2023 off to you blue to a participatory online Institute that is communicating learning and practicing applied active inference these are all of our social media links when you where you can find us if you're looking for us outside the industry this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we will be trying to follow a good video etiquette for live streams other than my cat who feels free to interrupt whenever she sees fit um you can reach out to us if you would like to get involved with any of our ongoing projects or live streams or propose a new project and today we are going to discuss uh this is bookstream number 001.06 so this is our sixth contextualization video or the contextualization video over chapter six um and we are going to study governing continuous transformation reframing the strategy governance conversation by Bijan kesri who was nice enough to join us and discuss some of his ideas with us last week this video is just an introduction to some of the ideas presented in the book it's not in a review or a final word in any way so we're just going to Jump Right In um and introduce ourselves so I am blue I am a researcher in New Mexico and I will pass it off to Daniel I'm Daniel I'm a researcher in California and looking forward to the continuation of the series Tyler hey my name is Tyler I'm a practitioner not a researcher so I lead product and protocol design at 4K protocol and I also do governance on the limits of Dow tooling with medigo all right blue awesome So today we're going to get into uh part two of the book so free energy governance and we're going to discuss the first chapter in part two um okay so part two free energy governments and we're discussing chapter six and there's kind of two sections and actually Jean broke chapter six for us so there's two sections of chapter six the upper echelon View and the predictive processing framework um and and here just a little intro to part two free energy governance um says free energy governance introduces a novel cross-hiorical generative framework that uniquely

applies the triplet of structure cognition and capabilities as the building blocks of the firm's generative model specifically in the context of continuous strategic renewal in distributed and discontinuous Market environments essentially six core theoretical perspectives together lay the foundation for this new conversation structure upper echelon View and prediction processing framework um so that's what we're going to discuss today and then I'm assuming that the next chapter is going forward are going to focus on cognition which encompasses attention-based View and an activist approach and capabilities encompassing Dynamic capability View and antagonistic neural networks perspectives structure cognition and capabilities form one integrated system under free energy governance failure in any one domain is likely to challenge firm survival the framework's core propositions outlined at the end of this chapter summarize free energy governance's ambition for a practice relevant theoretical contribution as both the generative logic of organizational governance as well as an early warning system predicting corporate decay um so Tyler do you want to jump in and read the absolute abstracts here View and prediction processing framework address free energy governance structural Dimension indeed upper echelon is the size strategic decision making however uev suffers from several limitations there's no explicit consideration for the board of directors and the language between the upper echelon and to the rest of the organization I.E the coupling of governance and operational challenges is unidirectionally top down and therefore at best and complete how the field data can systematically talk to Upper Echelon and influence their values procession and attention remains of black bugs it's prediction processing framework Embraces structure more holistically addressing the hierarchical disconnect and more importantly the circular causality between top down predictions and Bottoms Up stimuli I don't know just the second part here dissonance bearing messages sorry dissonance pairing messages occur at the firm stimuli level bottom close to its resource markets long before top leadership has any sense of a living challenge prediction processing framework is determined by three categorical imperatives first upper echelon's production models and predictions must be explainable second each of the hierarchical layers must have formal and informal interfaces coupling each layer to the next layer and third the firm's generative model is the firm's Eco Niche prediction processing framework provides an entirely New Perspective and language embracing strategic renewal as a matter of bottom-up top-down prediction error minimization not least setting 3 the full potential of the human machine interface I mean it's interesting here like I wonder what he means by human machine interface like humans as a functioning machine like um as an organism maybe or as like a um I I don't know or like directly like the

human computer interaction like how we interface with our technology so I I was curious like this I found this human machine interface um Clause kind of like threw me for a Loops a little bit you guys know any comments imagines a world where like you're very tied in with the the AI and I personally haven't like witnessed this myself like being in an organization where like that's like a seamless integration where like you might use like a machine learning framework but rather that like it is like a dialogue between the the frame ml model and then the human I think that's what he's kind of imagining um but I think it's a little bit unclear I think you're right blue so like I've I've like studied and taken like courses and intelligence with people who study like their course of study is like human computer interaction like the interface between humans and machines and like it's always like um you know that there are topics of study are very different than um like thinking about themselves as a firm right right like they they think about like their their topical study like interacting with like a wellness robot like you've seen like the health like little robot you know for elderly people that like checks up on them and it you know it's like the how how do elderly people best interact with computers and children and they have a very different level of interaction both like the older generation and kids they have different ways to interact and interface with um computers and like they do it best in an embodied kind of machine which I found super interesting like where you know in the mid-range we can just talk to the we have no problem talking to Alexa and it could just be like a microchip like speaker in the wall and it would be be a big deal but like they want something in a teddy bear or some kind of like machine that that like looks has a face or something like that and they interact with it in a much different way much much more efficiently actually so I found that super interesting but this is what I think about when I think about the human machine interface is like the actual like Markov boundary right like between humans and machines and and so like what is the potential of a Markov boundary I don't know anyway so sorry I'm rambling um okay so just a few keywords here upper echelon view prediction processing framework we'll go back and revisit it double learning look into optimization organizational sense making and um I think the the biggest chunk of new information that Bijan brought into this chapter what were these habits of mind um okay so do you want to read this for me Tyler upper echelon View cool so uev defines the organization as a reflection of its upper echelon the I.E its top management team UAV Central premise is that Executives experiences values and personalities affect their field of vision their one field of vision the directors they look and listen select a perception what they actually see in here and three interpretation how they attach meaning to what they see in here so essentially UE Embraces

sensing and sense making not only as an essential leadership qualities but as a foundation for strategic choices and firm performance cool and so there are a couple claims kind of associated with this upper echelon View um Bijan says however upper Ashland view suffers from several limitations in particular there is no explicit consideration for the board of directors as part of the upper echelon the many dimensions by which a non-executive chairman relates to the CEO for example are likely to determine strategic choices he goes on to say um well upper echelon view has neglected the board of directors free energy governance emphasizes the importance of the board as part of a firm's upper echelon top management tier and the board of directors here jointly labor labeled as the upper echelon are together the locus of strategic decision making more importantly the board of directors can potentially assume a unique cognitive role and strategy process in the form of dynamic board capabilities that is highly complementary to that of senior executive leadership strategic capacity and I was wondering Tyler maybe if you have any comments here on like you know normally the board is not included as the upper echelon and bishan is like making the case for their inclusion is that typical in like organizations or what you think uh no that that's not common and I think John like touched on this in our last call with him the last discussion is that like usually you see the board as more of like a sounding board as that there are a they give a final sign off a rubber stamp they make sure you're not doing anything totally illegal and irresponsible but they're not really there to like really be in the weeds with you uh whereas what Bajan is arguing is like no actually like they have by virtue of being kind of separate from the day-to-day work they can give you a more unbiased view of like what you should be doing strategically like the capabilities you should be building and that they actually are like a really integral part of your whole strategic process and so it's unusual to think of them as like these two like really critical halves usually people think of the board as this like nuisance that they have to appease not as like a real thing that can actually add value to the organization absolutely thank you for that input um okay so a couple more things to say about UAV and we just kind of excerpt it from the text um upper echelon view is far too restrictive and one-sided top-down to explain through the lens of leadership qualities the anatomy of success and failure of strategic renewal this book invites to a new stream of upper echelon youth Centric research that operationalize upper echelon View's impact on current performance not so much in terms of antecedents and consequences of Team diversity but as a function of structure cognitively public governance and operational channels in the form of top down bottom-up circular prediction error minimization second cognition embracing environment as a matter of action-

centric enactment and third capabilities developing Duality management as a dynamic board capability to rewire the plane while flying it um he goes on to say the prediction processing framework allows us to embrace structure more holistically I don't know if you guys have any comments here yeah let's just say that we're in a tripartite firm with the upper echelon consisting of the higher management and or the board of directors middle management and then the front lines as Vision mentioned last week overly emphasizing in a given Eco Niche any one of those three sections is going to lead to maladaptive decision making and implementation so the question that I see these repeated structure cognition capabilities returning to is about striking and adaptive balance amongst the upper echelon the middle management implementational layers and the front line day-to-day experience just like the body and the organism strike a balance between the primary perceptual organs the intermediate processing and then the central processing well thank you all right so moving on to the prediction um Tyler do you want to read this for us do you mind all right so neither governance or strategic strategic renewal research has addressed the hierarchical disconnect of the top down bottom-up circular causality sir then acceptable contributions aside uh ppf opens the pro perceptual black box of how shared meaning and expectations are constructed and optimized and pursued optimizing the performance sustainability of strategic renewal in management science PBS episode logical Origins can be traced to Argus agaris's theory of action perspective specifically the concept of double Loop learning defined as a form of learning that challenges the existing routines and Status Quo action-centric learnings at the heart of cross-cycle prediction processing framework there's a need for fundamentally for a fundamentally new organizing framework systems preparedness to unlearn Powers the receptivity the corrective feedback of the decision-making unit and hence is critical to ppf's top-down bottom-up learning prediction error immunization challenge well Daniel do you have any comments here yes um in Bayesian learning learning really is on learning you have to move the distribution around and so Bayesian learning isn't just throwing more papers onto a stack or onto a knowledge pile but when we have to update How likely we think things are that isn't unlearning so it's interesting to see how it's connected here to like openness to the unknown negative capabilities and the way in which in the single Loop which I know we'll get to you can Implement and do local maximization and maybe be locally adaptive but if we really want to pull back and amidst these discontinuities continue to find success then we need to engage in that kind of double Loop learning you totally took the words out of my mouth and actually just said it so much better so um I I also was like really reminded of Bayesian learning

here with with unlearning because like for all the reasons you just stated um okay so double Loop learning just to revisit I think we talked about this maybe not last week with bijanpa just a couple weeks ago um so double Loop learning occurs when in order to correct an error it's necessary to alter the governing values of the master programs so in this little figure it's just like single Loop learning is actions and that's the consequences actions assess the consequences where the double Loop learning zooms out to kind of see like what governing variables influence the actions that that are being taken um so it's kind of like the the setback in a hierarchical control model um and Bijan says that we need error feedback loops to adjust predictions and more importantly to eventually update prediction models um AKA double Loop learning okay so the Jean goes on to make the claim he says upper echelon Remains the centralized purveyor of predictions as well as Trustee of prediction models hence the prediction processing powered free energy governance framework is distributed at the bottom but centralized at the top however for prediction error minimization to work each hierarchical layer must be dedicated to one simple first order principle minimize free energy to minimize error and surprise consequently prediction models will generatively evolve by way of self-learning this is the essence of unsupervised self-organization um any comments here sure so in terms of communication what are these different parts of the um what are the different parts of the organization or organism sharing with each other so they might just be representing what they are perceiving like everyone is just filling others in on what they're getting so that's one imperative is just unbiased pass through a communication um another imperative might be give the other person what they expect so here the management is saying we expect that you're going to sell 10 cars and then the front line says yes we sold 10 cars where free energy minimization as a way to minimize or bound error and surprise comes in is that it says actually the communication imperative shouldn't be simple pass-through and it also shouldn't be conformist but rather what is communicated should provide information about to what extent the expectations of your communicating partner are being violated so if they say we expect 10 cars the minimum communication might just be yes if that's true and then maybe there's um a dial that can be turned depending on whether it's fewer than that or more than that and so that puts communication into an interesting setting where we're explicit about what the in this case top down predictions are and then we situate the bottom up communication as reflecting the adequacy of those expectations rather than just being plain pass-through or conformist yeah just to build on that a little bit because in my experience like there's like like you sit down like two kinds of communication there's one of like very clear like hey this is what we're expecting

did you meet that like a very clear prediction and there's another one that's more just like dissonance like things just like aren't working well and people don't fully understand why and this is my experience like really trying to communicate in a very like clear linear way like through each stack of the hierarchy uh about like trying to warn someone of a of an issue for example or communicate like the new world or new like how you should update your generative model that tends to not work nearly as well as just things not working well and people bumping up against uh discontinuities and dissonance themselves and people really like oh there's like a problem in the system this is a signal that we need to fix it and usually that's in my experience like how communication happens between hierarchies and it feels really bad and it feels very chaotic but at the end of the day I think that's how like the organism of the firm processes information yeah it's asking too much for the skin cell to provide a situational report on the wound all it can and must do is report a difference from expectation that gets clustered with other incoming sensory modalities and integrated by the upper echelon to make a decision about okay now we're going to get a Band-Aid so understanding that the skin cell is not going to tell you to get a Band-Aid but also the Band-Aid decision-making unit isn't going to have direct access to the skin cannot be implemented at the firm level how like that that coupling between different different layers of the hierarchy man can I talk today the coupling between the different layers of the hierarchy how should that work so like just for example like the skin cell doesn't necessarily need to know what is happening in the liver right like it's not like a direct coupling there um and and so how how do you selectively choose like it as like a management like what to communicate down the chain like if I'm in the upper echelon like what does the board communicate to the top management here what is the top management tier then communicate like Downstream and how do you filter that information so that it's like like how do you like prioritize salience here and just um not like open up a fire hose like here's all the information because that's not like effective for communicating but it also like you like that disconnect like any kind of filtering can lead to like a level of Disconnect that's maybe unwanted so how do you like prioritize that information flow um down layers of the hierarchy and like I um always am interested in this so like whether it's biological or organizational I have a couple examples to like illustrate that so for example I worked at a company called actually I won't name any names here but it was a tech company and it was like a very it was probably like 400 people and it was like structured very hierarchically and so basically from the bottom bottom rungs to the top you would have each person would report up to their unit and have like hey this is what we did

these are like risks you should be aware of right and this would happen on a weekly basis and then that would get bubbled up into the next running out of hierarchy which would be it keeps getting synthesized and refined at each level of the hierarchy until you get up to like the CEO right which is presume like ostensibly like uh condensing all this information from 400 people up to like one report but the CEO gets of like five bullets of takeaways right and so like what happens like in Bajan even talks about this in the chapter is that like it gets so filtered by that point and also each rung in a hierarchy has like their own incentives and things that they're trying to like politically maximize for that becomes like not even that useful so what what's a lot more useful though is when things uh well I guess two things one is if things go wrong like say there's a bug in the app and something needs a firing to be put out right away that's information that skips or all the wrongs and goes straight to the CEO another one is when people would a lot of times Executives would schedule meetings but just random people in the organization so you'd have like an executive who's like the CFO would schedule a meeting with some engineer and just to like basically cross the chasm of all the different levels of the hierarchy just be like hey like what's going on like tell me like what you perceive the problems to be in your sub unit and that's a way for them to kind of get like this unfiltered view of like random people like by sampling random people across the organization so I think they're like ways you can do this but yeah it's really it's really difficult foreign okay so moving on we'll talk a little bit about optimization and um I just yeah put up here like the um you know distributed versus um hierarchical levels the networks um okay so Bijan says the challenge is not only one of building interfirm connections between upper echelons top-down predictions and field Management's bottom-up stimuli but optimizing the top-down bottom-up process within each and every layer of why am I fixing that of the hierarchy in form of feedback connections optimization is Central to prediction processing framework this optimization makes every level in the hierarchy accountable to others while the generative sense making processes and capabilities are critical performance element at every level of the hierarchy the efficacy of such capabilities is determined by top-down belief propagation and bottom-up message passing within and across organizational layers or business units um yeah so I think like that optimization for like maybe information or I don't know like what are you optimizing optimizing the connection optimizing the flow of information optimizing um the link between up and downs like like what are we really optimizing four I don't know yeah it's it's a little bit vague in how he talks about it there's obviously something I got kind of a little bit frustrated with because like actually building that top down bottoms up kind of connective tissue is like a

huge area of research and there's like all these like different um approaches for it it's like a really famous one is holocracy that I'm not sure you guys are familiar with holography hypocrisy he got famous because it was implemented by by Zappos uh to like mixed results potentially but what a cool way that they do it is that it's not like a very traditional pyramid hierarchical structure but each like subunit will kind of dynamically form bonds with other levels of the hierarchy kind of arbitrarily depending on needs as they arise so like but they have a whole system for how those uh those bonds conform and how you generate links across different levels of the hierarchy and so that's just to say like this is like a huge area of like research and experimentation that no one really has like a great answer for um and yeah it's this is like a huge open-ending questions there's no General strategy for a bug so it doesn't seem like there's going to be a general organization chart or communications platform and technology is changing so rapidly and intelligence augmentation this is helping us like sketch the first lines on our notebook that takes us into a discussion and an Adaptive process and that is the reframing of the strategy governance conversation it's not this is the strategy Playbook it's not the um homework problems but it's it's bringing some ideas to the table that are mixing it up so like I don't know I think about optimizing like sense making at each level um because of like the quote that's stated here and so like how do you like optimize sense making across layers like it definitely has to do with like information prioritization I think um but but it's like how to selectively choose like I guess it's like a very open-ended question but I don't know it's like I would like some concrete examples it's something like I would like to drill Bajan on like how have you ever seen this done and like what worked and what didn't um for as far as optimizing like this hierarchy flow as to what is being optimized one sort of capital maximalist approach is like profit is to be maximized or risk discounted profit or a triple bottom line like there are imperatives for the organization to optimize but that doesn't mean that that becomes the imperative at every communication step and kind of the dream or direction of informational free energy minimization would be you'll be doing as good as you can in the niche from like a energetics and Metabolism perspective when you're informationally optimizing if you're a bear making the best possible decisions you will survive in the niche and so it's kind of like recognizing the difference between the ecological success of an organization or or organism recognizing that that embodied success and death isn't the informational imperative but for things that work and not things that don't fulfilling the informational imperative should lead to the embodied success because it's defined in that way so it's easy to turn to like maximizing profit as a firm objective but I think that like that is

short-sighted like I definitely think that there's many other things like if you're making the most possible money of producing light bulbs but right like so say I'm making a light bulb and I'm like maximizing profit on the light bulb and I'm I'm producing a million of them for pennies right and they're selling for you know ten dollars each or whatever um but I'm like polluting a river because I'm making the light bulbs so cheap like I'm not like adequately disposing of waste or whatever then like that's gonna have blowback like later either through like EPA or through like customers saying I don't want that because it's not sustainable or whatever so like if you're just exploitative or yeah exploitative um then it's not gonna work out in the long run so like maximizing profit is a very like delicate thing like especially in terms of like company persistence and sustainability like sustainability not like an eco-friendly way but like sustainability of the company itself yeah I mean this oh sorry I just I was thinking of fermenting bacteria initially certain strategies work but then that can lead to the production of metabolites that that strategy will no longer be optimal in so one has to consider like externalities in the niche and of course all of this is just a statistical framework that's way before we get into ethical imperatives or preferences yeah I mean so I think there's something I think we covered in the the third live stream but it's like okay how do you how do you heard cats basically and get all these different levels of the organization to move in the same direction and kind of balance that these different things you might want to optimize for that in the long term maybe it goes for profit but you might have like sub problems that you need to solve that are not strictly profit and so one of the most famous ways of doing this is okrs so like objective key results and so like at your top level of the organization maybe it is profit or maybe it is like some kind of double bottom line but then as you go into like subunits of the organization like say you have customer service because your customer service team is not optimizing for profit they're optimizing for like a customer experience or like csat for example right and your legal team is not optimizing for profit they have like their own measure that they're they're optimizing for the number of cases they win or legal risk you know avoided I don't know how you quantify that um but there are ways that you can kind of break the bigger the problem of profit into sub problems uh and rather than having you know everyone going towards profit at the same time all right oh so perfectly perfect segue and so organizational uh sense making so I'll read this organizational sense making is not about optimally approximating and objectively defined external reality but developing and enacting models of Reality by way of top down bottom-up predictive coding effectively the future arrives in the present and becomes the Past Times time moves at different rates in the brain's hierarchy as is

true for the firm while time has strictly no speed humans project speed onto time as relations and interactions increase in intensity and complexity the firm's very existence is determined by its curiosity field interaction with its external environment fast at the reflex low level I.E resource resource markets and more slowly at higher cognitive levels I.E strategic decision making I mean this you know reminds me of um like a paper that Daniel and I have been working on and just how how temporally active inference and fep operates at different um fast and slow is it kind of reminds me of how and pretty much every company I think I've ever worked at there's always this tendency for the Bottom Rung of the organization to think that the top round are just like a bunch of like slow bumbling idiots there's like this temporal disconnect of like the Bottom Rung of the organization is giving feedback immediately and like they're running little tiny experiments every single day where they're getting immediate feedback whereas like the top wrong is like you know the board maybe they're meeting quarterly and then they have like some mandate and they don't really get feedback synthesized to them until like another three months uh and so like there's always this tendency for like there's always this temporal disconnect and it always creates this kind of like boundary or disconnect between the bottom and top of the this point of time having no speed is quite interesting we talk about oh it's the food is taking it's too slow or slower time scales faster time scales longer distances like it feels natural to talk about the rate of speed but even without going into too much uh reconceptualizations of speed it just isn't necessarily that way and then in the brain as to like time we could think about a neural firing pattern that takes place over some number of milliseconds and then there's slower waves and rhythms they compose and they influence each other they're all happening at once technically it's all happening in the same way at the same place the same exact input outputs but still it makes sense to talk about some of these slower or larger waves and some faster waves like in the EEG rhythms so striking a balance between the everything all at once real time unfolding and then these day week month year decade nested time scales is a big part of sense making definitely okay habits of mind um so I I liked this uh because I've read good to great um the Jim Collins book and when Tyler and I were first starting out the series we talked about like how that's very like leader-centric and very like top down like the the ideas that are presented in that book um and so I I was like excited to see this book uh good to Great to gone by Alan werzel I have not um read it but he talked about companies like I think specifically like Circuit City is the company right oh yeah it's here on the cover of the book but Circuit City was like one of the companies that uh Jim Collins went through and good to great and then now like there is no more Circuit

City and so he I guess kind of outlines like the failings of like what makes a good company a great company and now like they're obsolete um and maybe why they're Obsolete and so he um outlines these habits of mind um and I don't I haven't read the book and I don't know enough but they're listed here if you want to um read them and maybe read the book um but these habits of mind that are are like necessary to um stay relevant maybe um so uh Bijan says that habits of mind May equally explain why upper echelon is generally resisting strategy process as a matter of top top down bottom-up prediction urbanization um and he goes on to make the claim that five out of where it's those 12 habits of mind are foundational to prediction processing framework fostering success of strategic renewal um and he briefly summarizes these habits uh Tyler do you want to read these scores did you find sure so first one be humble run scared so continuously doubt your understanding of things business success contains the seeds of its own destruction two curiosity sustains the cat the world is constantly changing be open and curious and strive to learn from others three evidence Trump's ideology and business as in politics decisions are too often based on unproven assumptions about what works and what doesn't we all need operating assumptions about human nature the economy and the like but when things do not work out as planned we need to determine whether our assumptions were based on evidence or ideology four confront the brutal facts the worst person you can fool is yourself ignoring or denying reality does not help it go away five chasing possible dream do not be limited by the tyranny of the war foreign so I thought this was cool I think a lot of these habits of mine speak to the ability to like rewire the playing while flying it um I don't know like what what Bajan stressed earlier in the chapter do you guys have any um additional comments here um okay so another claim and this might be who might be oh yeah that's getting close to the end okay Bichon says it is the circular causality of belief belief propagation message passing where prediction processing Frameworks Effectiveness and the organizational context most likely breaks down neither the bottom-up message passing nor the top-down belief propagation are unobstructed we fail to appreciate how organizational friction which he defines as high levels of free energy compromised firm performance due to First top-down enforced predictions and prediction models that are not receptive to bottom-up challenge second leak top down bottom-up lateral generative sensing and sense we can capabilities in the form of error signaling and processing and third poor communication and absence of a language that is suitable to level leverage prediction processing framework for strategic renewal um so I thought that uh that was kind of neat um like how how the these are the faults right um and then he goes on to say in the

organizational context prediction processing framework is determined by three categorical imperatives and I've abbreviated these kind of just shortening them so that they fit on the slide first upper echelon's prediction models and corresponding predictions must be explainable and explicitly articulated and um yeah so that goes into the communication salience prioritization he says the language of communications layers are critical to Resource configuration from performance and eventually firm survival um he also says each of the hierarchical layers must have formal and informal interfaces that couple each layer to the next layer for a given hierarchical structure of a firm to have any legitimacy generative top-down bottom-up sensing and sense making at every hierarchical level should equally make sense before an error message travels to the next level it should not and cannot be any bypassing of hierarchical layers um I guess that's like don't go over someone's head um but also like yeah I I wonder like about the bypassing about like the information flow and the throughput and how big does that channel have to be um and lastly he says the firm's when we perceive the world as uncertain and unpredictable it is a mere reflection of the inadequacy of our generative reality model um so I think that that's how to avoid the errors mentioned in in the previous slide any comments here final thoughts on the material one thought is I like number one where he's talking about how you have to articulate and communicate top down like what your actual model is so it can be challenged and something that I've noticed is that oftentimes like people will communicate their predictions but not necessarily have those predictions were generated or the components of that like the components of the assumptions that went into that prediction and that makes it really hard for other levels of the hierarchy to like even know what was going on and what they should be challenging at all and a lot of times you'll just assume like okay like they probably know what they're talking about like they're the experts so like let's just hope for the best um but I really like that he said like no you have to explicitly say what your model is so it can be challenged to the going over someone's head on a previous slide it said organizational friction is High free energy we might want to finesse that by saying relatively High free energy because in general we're not interested in the absolute value of a free energy calculation but rather the gradients and the direction at which we could act to improve or increase it so it's not about the total amount it's about the relative amount so let's just say it turned out that somebody from an org chart perspective did go over their head that could be understood as a policy selection authorized or not to structurally rewire the communications architecture of the firm now that rewiring in a situation may be associated with a decrease in organizational friction and therefore a decrease in the free energy so it may

be an Adaptive decision because no org chart is the final word but that doesn't mean that every time somebody tries to rewire the communications architecture it would decrease friction if people were just getting all these side messages and there'd be so many issues as well so it's it's not to say to do or not to do it's giving the areas of consideration the structure cognitive and capacity oriented aspects that we can have that ongoing conversation and some principles that help us take selection and still do things in an uncertain world with this little fun caveat that the perception of the world as uncertain and unpredictable is actually an inadequacy of our generative reality yeah I mean even skipping different levels of the hierarchy could be something that you explicitly plan for as well and build processes around so for example I worked at a company that they call it like the big red button whereas if you were at the bottom wrong in the organization and you saw something that was going terribly wrong wrong you could basically push this big red button and it would file this like report directly to the C-suite and would skip all the levels and then they would follow up with you directly and so like they explicitly said like hey we know that they're going to be things that are shielded from our view and we want to know about it and they build a process explicitly around it that so in terms oh sorry go ahead Damian that reminds me of cells signaling and then the ultimate red button is like when the cellular contents are spilling out it's like my only message to share is that I am no longer and all my insides are like spilling onto you so it's like if there's a protein that's usually internal and it's being you're being bathed in it that's the red button that you needed so yes that kind of emergency or a conditional rewiring maybe there's many patterns and motifs where we might want to embed that kind of flexibility into the communications architecture so even though it is a dynamic Communications architecture at a higher level it's actually not changing in its Dynamics like different people are going to be in different video calls but that doesn't mean that the communications architectural plan is changing it just means that difference individuals are connected differently so in terms of um sharing the model Tyler like when you were talking about um you know sharing how like the the top level will share their predictions but not necessarily how those predictions were generated I'm like really reminded of um academic research in which it's I feel like everyone is so like shielded from what's going on like you have the principal investigator they like write a grant the grant has like three specific games it's just an example so then like those three specific games the each grad student will be working on one of the Apes right or maybe they'll have a team working on achieving one of the three specific aims right and like these teams are doing totally like work that they think is totally unrelated like they have no idea like the overarching

process of the grant like I wonder how often like the pi shares like the entire Grant with every student like I mean that's not something that I typically see and I think like that there's a lot of shielding going on and I don't I don't really know why like is it is it like because it's just too much information or because like they just want the focus on to be on that one thing like they want this the students to have this myopic view or and I I just understand you like what is your experience there like do you think that like pis generally share the entire model or they just share a piece of it like what is your experience and do you think that that makes research dysfunctional interesting I've seen some investigators who shared the grants or other ongoing strategy documents or not it reminds me of a computer program like if you could perfectly isolate every variable and for Loop and you can really clearly say this is the scope of this script and all of that that may work in a computer system but humans are always curious about the context and the past and the future and Alternate futures that kind of strict information encapsulation approach might increase overhead decrease context have to spend a lot of time re-explaining things it's not that one strategy is going to be in general better than the other but it seems like it could be a better professional conversation if the lab strategy governance were framed in a way that accounting for people's differences in like familiarity with grant writing or with experiment design at least giving lob level awareness doesn't seem like it'd be a bad thing yeah there's an analogous problem within a business organization and some of you type like okr is where you take it like this big goal you had like let's say it's growth and you split up into subcomponent parts so one problem I've had in one organization actually multiple organizations is on the growth funnel so like how you get leads and then how you convert them into like actual conversions and sales right and so this what often happens is that there's a trade-off between getting more leads and the quality of those leads and so what you'll do is you'll have like a top level growth initiative like we have to grow by 10 and then you're like cool like marketing people increase your leads and then uh product people all right increase the conversion on this funnel so but what actually happens though is if the marketing team increases their leads they often get more poor quality leads like you're dipping into like lower quality Elites increase your leads and so that actually reduces the conversion of the next step of the funnel and so like they're both narrowly optimized on like their one little piece of the puzzle but they actually really directly impact each other and then you get into these like really painful conversations between your your marketing team and your your growth product team which is like really like if they were just had this like more holistic view to begin with like no we're just trying to grow guys like that's what we're trying to do like we don't care

about your little piece of the puzzle and so like there's often this time where like by breaking things off too narrowly uh you really missed a bigger picture and this is super super common the last we'll have it having the um the apriori partitioning means that you are saying like uh not even the structure of the problem is going to change again if you only have to partition the problem one time then maybe it can be cut and dry but those kinds of dependencies are why we need uh these kinds of composable partitioning strategies so that different parts of a base graph can be identified because we're looking at a lot of text but all of these Notions of a generative model generative process we can also see that kind of a base graph representation in the background especially as these types of perspectives move closer to application awesome well that's the end of the content here for chapter six we will delve into chapter 7 next week unless you guys have any final things to say um that's it for me so thanks for for being here and I'm looking forward to carrying on pushing forward learning more about free energy governance and future discussions with the author thank you all right thank you until next time bye